

Extrapolation and Validation Design Responses

Verbatim transcript of two prior responses: the proposed uncertainty-tolerant design and its comparison with the current implementation.

Response 1: Improving extrapolation and validation

The key is to validate transparency and reasoning discipline—not require every recommendation to appear verbatim in one paper.

Your system should explicitly allow synthesis across sources while forcing the model to distinguish:

- What the literature directly reports
- What is transferred from an analogous system
- What the model inferred
- What remains unknown
- What must be experimentally validated

A useful principle is:

Uncertainty is permitted. Undisclosed uncertainty is not.

1. Give each recommendation an evidence status

Replace a simple “reported or inferred” distinction with a more expressive classification:

- **directly_reported**: The source tested essentially the same protocol and relevant conditions.
- **adapted_from_analogue**: The source differs in chemistry, format, temperature, loading, or another important characteristic, but the model provides a scientific transfer rationale.
- **triangulated**: No paper reports the complete recommendation; multiple sources jointly support its components.
- **hypothesis**: Mechanistically plausible but weakly supported by the retrieved literature.
- **insufficient_evidence**: The system cannot responsibly propose a value or strategy.

This lets the model make useful connections without presenting them as established findings.

2. Require an explicit evidence bridge

For every adapted or triangulated suggestion, require the model to explain the connection:

```
{
  "strategy": "Use a multistage constant-current protocol with a reduced upper-SOC current.",
  "evidence_status": "adapted_from_analogue",
  "evidence_chunk_ids": [
    "li2024:chunk::0012",
    "smith2023:chunk::0008"
  ],
  "source_conditions": {
    "chemistry": "graphite/NMC811",
    "form_factor": "pouch",
    "temperature": "25 °C"
  },
  "target_conditions": {
    "chemistry": "graphite/NMC622",
    "form_factor": "cylindrical",
    "temperature": "25 °C"
  },
  "transfer_rationale": "Both systems use graphite anodes and layered-oxide cathodes, so the reported upper-SOC polarization trend is potentially transferable.",
  "important_differences": [
    "The cathode stoichiometry differs.",
    "The cylindrical cell may dissipate heat differently.",
    "Electrode loading and electrolyte formulation are unknown."
  ]
}
```

```

],
"assumptions": [
  "The target cell has comparable graphite fast-charge limitations."
],
"confidence": "low",
"required_validation": [
  "Begin below the reported peak current.",
  "Monitor temperature and voltage relaxation.",
  "Check for lithium-plating indicators before increasing current."
]
}

```

The “evidence bridge” is the important part: source → similarity → difference → inference → validation requirement.

3. Prompt for bounded extrapolation

The system prompt should affirmatively permit synthesis:

You may construct a protocol by combining evidence from multiple studies or transferring findings from analogous cells. Do not require a complete protocol to be reported in a single source. Clearly label every recommendation as directly reported, adapted from an analogue, triangulated, or hypothetical.

Then establish boundaries:

For adapted recommendations, identify the relevant similarities and material differences between the source and target systems. Explain why transfer may be reasonable and how the differences could invalidate it. Never describe an inferred value as experimentally demonstrated for the target cell.

This is better than telling the model simply to “only use supplied evidence,” which can make it either refuse too often or disguise inference as retrieval.

4. Treat numerical extrapolation more strictly than qualitative extrapolation

Not all inferences carry equal risk.

Qualitative synthesis might reasonably conclude:

- Reduce current at high SOC
- Use temperature-aware current limits
- Consider pulse or multistage charging
- Monitor polarization or relaxation

Numerical synthesis is much riskier:

- Exact C-rate
- Exact voltage limit
- Exact SOC transition
- Exact pulse duration
- Exact temperature cutoff

The validator should allow qualitative transfer relatively freely when disclosed. Numerical values should require one of the following:

- Direct support under sufficiently similar conditions
- A cited calculation or explicit scaling rule
- A conservative bound derived from evidence
- A clear label that it is a starting hypothesis for controlled testing

If none applies, the model should return a range or say that the value cannot yet be determined.

5. Make confidence multidimensional

A single confidence label can hide important differences. Score at least:

- **evidence_strength**: How directly do the papers support the recommendation?
- **transferability**: How similar are the source and target systems?
- **completeness**: Are important operating conditions known?
- **agreement**: Do multiple sources support or contradict the recommendation?

For example:

```
{
  "confidence": {
    "evidence_strength": "medium",
    "transferability": "low",
    "completeness": "low",
    "agreement": "medium",
    "overall": "low"
  }
}
```

This communicates more than an unexplained “low confidence.”

6. Validate disclosure rather than scientific certainty

Your deterministic validator cannot reliably determine whether a scientific analogy is correct. It can enforce whether the analogy is auditable.

For an adapted recommendation, validate that:

- At least one real retrieved chunk is cited.
- The source conditions are stated where available.
- The target conditions are stated.
- Important differences are acknowledged.
- A transfer rationale is provided.
- Assumptions are explicitly listed.
- Unsupported numerical precision is flagged.
- Experimental validation requirements are provided.
- The wording does not falsely say the target protocol was demonstrated.

A missing disclosure should produce `pass_with_warnings` or request a repair—not necessarily reject the entire answer.

Reserve rejection for serious failures such as:

- Fabricated citations
- No evidence retrieved
- Claims of direct experimental support that the evidence does not contain
- Unsupported safety-critical numbers presented as established limits
- Recommendations that exceed an explicit manufacturer or source constraint
- Invalid or internally inconsistent output

7. Let the agent retrieve analogues deliberately

When an exact match is unavailable, the agent should broaden retrieval systematically:

1. Search for the exact chemistry, format, and conditions.
2. Identify which dimensions are unsupported.
3. Relax one dimension at a time:
 - Same chemistry, different format

- Same anode or cathode family
 - Same degradation mechanism
 - Same temperature regime
 - Same charging-control strategy
4. Retrieve neighboring chunks for experimental context.
 5. Compare supporting and conflicting evidence.
 6. Construct the recommendation and disclose the transfer.

This is more defensible than issuing a single broad search and asking the model to improvise from whatever appears.

8. Present protocols as proposals, not prescriptions

The final interface should use language such as:

- “Evidence-supported starting protocol”
- “Proposed initial test envelope”
- “Adapted from analogous systems”
- “Requires cell-specific validation”

It should avoid implying that the application has produced a ready-to-deploy controller setting.

A strong final answer could say:

No complete protocol was found for the target NMC622 cylindrical cell. The proposed upper-SOC current reduction is adapted primarily from Li et al.’s NMC811 pouch-cell study. Transfer is plausible because both systems use graphite anodes and layered-oxide cathodes, but differences in thermal behavior, cathode composition, loading, and electrolyte formulation reduce confidence. The suggested values should therefore be treated as conservative experimental starting points rather than validated limits.

That is useful, evidence-based, and honest about what the literature does not establish.

For your implementation, I would modify the output schema, system prompt, validation rules, and UI together. The most important change is replacing binary “supported/unsupported” validation with a provenance-aware ladder of direct reporting, adaptation, triangulation, hypothesis, and insufficient evidence.

Response 2: Differences from the current implementation

The current implementation already supports inference in principle, but it does not require the model to explain or structure that inference sufficiently.

What is already implemented

The current system prompt says:

- Use retrieved evidence before answering.
- Cite every protocol suggestion.
- Distinguish reported, synthesized, and inferred recommendations.
- State missing conditions and conflicting evidence.
- Never invent numerical operating or safety limits.
- Do not represent a literature-derived protocol as deployment-ready.

The current validator checks:

- Every recommendation cites at least one retrieved chunk.
- Every cited chunk actually exists in the retrieved evidence.
- `reported_or_inferred` is one of the permitted categories.
- Confidence is low, medium, or high.
- Numerical values appearing in the recommendation also appear verbatim in the cited passages.
- Invalid citations and malformed output cause rejection.
- Unsupported numerical values cause `pass_with_warnings`, not rejection.

So the system is not currently restricted to protocols directly reported by a single paper. It can already produce a synthesized or inferred recommendation.

What is missing

The main weakness is that the current schema only requires this:

```
{
  "reported_or_inferred": "inferred",
  "applicable_conditions": [],
  "rationale": "...",
  "limitations": [],
  "confidence": "low"
}
```

That is too coarse. A model can label something “inferred,” provide a vague rationale, and technically pass validation.

The current system does not require it to identify:

- Which source system it is transferring from
- How the source and target chemistries are similar
- How they differ
- Why the difference might or might not invalidate the transfer
- Which assumptions are being made
- Whether the recommendation combines multiple papers
- What experiment would validate the extrapolation
- Whether low confidence comes from weak evidence, poor transferability, incomplete conditions, or conflicting studies

Direct comparison

Area	Current implementation	Proposed design
Evidence classification	reported, synthesized, inferred	directly_reported, adapted_from_analogue, triangulated, hypothesis, insufficient_evidence
Cross-chemistry transfer	Permitted implicitly	Permitted explicitly, with a required transfer rationale
Source conditions	Optional free-form text	Structured source chemistry, format, temperature, and other conditions
Target comparison	Not required	Explicit source-versus-target comparison
Important differences	May appear in limitations	Required for analogue-based recommendations
Assumptions	No dedicated field	Required structured field
Validation plan	Not required per recommendation	Required for adapted, triangulated, and hypothetical suggestions
Confidence	One unexplained label	Evidence strength, transferability, completeness, agreement, and overall confidence
Numerical extrapolation	Warns if a number is not verbatim in the cited chunks	Distinguishes reported values, conservative bounds, calculated values, and speculative starting values
Validator	Checks structure, citations, labels, and numbers	Also checks whether inference is transparently justified
UI	Shows strategy, rationale, confidence, and citation IDs	Would display evidence status, transfer rationale, differences, assumptions, and validation needs

The largest practical difference

Currently, this could pass:

Use a staged protocol for this LFP cylindrical cell because a cited NMC pouch-cell study found staged charging beneficial.

It has a citation, an inference label, a rationale, and possibly low confidence.

Under the proposed design, it would need to say something closer to:

This is adapted from an NMC pouch-cell study, not directly demonstrated for the target LFP cylindrical cell. Transfer is based on the possibility that both systems experience increasing anode polarization at high SOC. However, cathode behavior, cell geometry, heat dissipation, electrode loading, and electrolyte formulation differ or are unknown. Therefore, only the qualitative current-reduction strategy is transferred; the source paper's exact C-rates are not. Cell-specific testing is required.

The proposed system does not necessarily make recommendations more restrictive. It makes extrapolation more visible, structured, and auditable.

One current rule may be too blunt

The numerical validator looks for the same numerical text verbatim in the cited chunks. That is useful for catching invented numbers, but it cannot properly accommodate:

- Unit conversions
- Values calculated from reported quantities
- Ranges synthesized from several papers
- Conservative reductions from a reported limit
- Interpolated experimental starting points

It currently issues only a generic warning for these cases. A better schema would require each important number to carry an origin:

```
{
  "value": "2C",
  "basis": "conservative_derivation",
```

```
"derivation": "Reduced from the cited 3C protocol because the target cell's thermal behavior is unknown.",  
"evidence_chunk_ids": ["li2024::chunk::0012"]  
}
```

The validator could then distinguish an explicitly derived number from an unexplained invention.

In short: the current implementation establishes a good safety baseline and already allows synthesis. The proposed change would primarily make analogue transfer and cross-paper reasoning explicit enough that users can understand exactly where the evidence ends and the extrapolation begins.